Chapter 2: Estimation and Losses

Part III — Statistical Foundations of Deep Learning & Reinforcement Learning

Omid Safarzadeh

Abstract

Every loss a network minimises is a likelihood in disguise, and every regulariser is a prior. Squared error is the negative log-likelihood of Gaussian noise, cross-entropy is the negative log-likelihood of a categorical label, and once you see losses this way the choice of loss stops being arbitrary — it is a claim about the noise. The same move explains regularisation: adding a penalty is multiplying the likelihood by a prior and maximising the posterior instead, with the L_2 penalty a Gaussian prior and the L_1 penalty a Laplace prior whose sharp peak at zero is exactly why lasso gives sparse weights. Threading both is the *score function*, the gradient of the log-likelihood, whose expectation is zero at the truth — the quantity estimation, and later policy gradients, are built on.

Contents

1 Losses are negative log-likelihoods 1

2 Regularisation is a prior 2

3 The score function 2

1 Losses are negative log-likelihoods

A probabilistic model says the data were drawn from $p(y \mid x; \theta)$. The **maximum likelihood** estimate maximises $\prod_i p(y_i \mid x_i; \theta)$, or equivalently minimises the **negative log-likelihood** $-\sum_i \log p(y_i \mid x_i; \theta)$. Different noise models give different losses — and they are the losses deep learning already uses.

Theorem 1.1 (Squared error is Gaussian MLE)**.** If $y = f_\theta(x) + \varepsilon$ with $\varepsilon \sim N(0, \sigma^2)$, then the negative log-likelihood is

$$-\sum_i \log p(y_i \mid x_i) = \frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2\sigma^2}\sum_i \big(y_i - f_\theta(x_i)\big)^2.$$

Only the last term depends on θ, so maximising the likelihood is exactly minimising the sum of squared errors.

Proof. Substitute the Gaussian density $p(y_i \mid x_i) = (2\pi\sigma^2)^{-1/2}\exp(-(y_i - f_\theta(x_i))^2/2\sigma^2)$, take logs, and sum. The θ-free terms are constant in the optimisation, leaving the SSE scaled by $1/2\sigma^2$. □

The left panel of the figure makes the point concretely: the sum of squared errors and the Gaussian negative log-likelihood, plotted against a candidate slope, are the *same* curve up to an affine transform, with the same minimiser. The identical argument with a categorical likelihood gives **cross-entropy**: $-\sum_i \log p_{\text{model}}(y_i \mid x_i)$ with p_{model} the softmax output is the

multinomial negative log-likelihood (Part II, Chapter 8). The loss is never arbitrary — it names the noise.

2 Regularisation is a prior

Maximum likelihood overfits when the model is flexible. The Bayesian repair is a **prior** $p(\theta)$: instead of maximising the likelihood, maximise the posterior $p(\theta \mid \text{data}) \propto p(\text{data} \mid \theta)\, p(\theta)$ — **maximum a posteriori** (MAP) estimation. Taking logs turns the product into a sum, so MAP minimises

$$\underbrace{-\sum_i \log p(y_i \mid x_i; \theta)}_{\text{loss}} \; \underbrace{-\log p(\theta)}_{\text{penalty}}.$$

The penalty *is* the negative log-prior.

Theorem 2.1 (Ridge is a Gaussian prior)**.** A Gaussian prior $\theta \sim N(0, \tau^2 I)$ contributes $-\log p(\theta) = \frac{1}{2\tau^2}\|\theta\|^2 + \text{const}$. Combined with the Gaussian likelihood of Theorem 1.1, MAP minimises $\frac{1}{2\sigma^2}\sum_i (y_i - f_\theta(x_i))^2 + \frac{1}{2\tau^2}\|\theta\|^2$, which is ridge regression with $\lambda = \sigma^2/\tau^2$.

Proof. The log of $p(\theta) \propto \exp(-\|\theta\|^2/2\tau^2)$ is $-\|\theta\|^2/2\tau^2$ up to a constant. Add it to the negative log-likelihood and multiply through by $2\sigma^2$: the objective is $\sum_i (y_i - f_\theta(x_i))^2 + (\sigma^2/\tau^2)\|\theta\|^2$, the ridge objective with $\lambda = \sigma^2/\tau^2$. □

Swap the Gaussian prior for a **Laplace prior** $p(\theta) \propto \exp(-\|\theta\|_1/b)$ and the penalty becomes $\|\theta\|_1$ — the lasso. The middle panel shows why the two behave differently: the Laplace density has a sharp peak at zero, so its MAP estimate is pulled onto zero for any weight the data do not strongly support, giving **sparsity**; the smooth Gaussian only shrinks. The right panel is the geometric version — the L_1 "ball" is a diamond, and the squared-error contours first touch it at a corner, where a coordinate is exactly zero.

3 The score function

Both sections rest on one object: the **score**, the gradient of the log-likelihood with respect to the parameter,

$$s(\theta) = \nabla_\theta \log p(y \mid x; \theta).$$

The MLE is where the total score vanishes, $\sum_i s(\theta) = 0$, and the two facts that make the score central are that, at the true parameter, its expectation is zero and its variance is the **Fisher information**.

Theorem 3.1 (The score has mean zero)**.** Under regularity conditions, $\mathbb{E}_{y \sim p(\cdot;\theta)}[s(\theta)] = 0$.

Proof. $\mathbb{E}[s(\theta)] = \int \big(\nabla_\theta \log p\big)\, p\, dy = \int \frac{\nabla_\theta p}{p}\, p\, dy = \nabla_\theta \int p\, dy = \nabla_\theta 1 = 0$, using $\nabla_\theta \log p = \nabla_\theta p / p$ and swapping gradient and integral. □

For the Gaussian mean, $s(\mu) = (y - \mu)/\sigma^2$, and $\sum_i s(\mu) = 0$ gives $\hat{\mu} = \bar{y}$ — the sample mean is the score equation's solution. This is not a curiosity: the identity $\nabla_\theta \log p = (\nabla_\theta p)/p$ is the **log-derivative trick**, and it returns in Chapter 7 as the engine of the policy-gradient estimator, where the "likelihood" is a policy and the score is differentiated through sampled actions.

Summary

A loss is a negative log-likelihood: squared error is the NLL of Gaussian noise, cross-entropy the NLL of a categorical label, so the choice of loss is a claim about the noise. A regulariser is a negative log-prior: MAP minimises loss + penalty, with the L_2 penalty a Gaussian prior ($\lambda = \sigma^2/\tau^2$) and the L_1 penalty a Laplace prior whose peak at zero yields sparsity. Both rest on the score $s(\theta) = \nabla_\theta \log p$, whose expectation is zero at the truth (Fisher information is its variance), and whose log-derivative identity $\nabla \log p = \nabla p / p$ reappears as the policy-gradient estimator in Chapter 7.